

Abhinav Shivkumar Ovhal

AI & Machine Learning Undergraduate | Full-Stack Developer
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PROFESSIONAL SUMMARY

Artificial Intelligence and Machine Learning undergraduate with hands-on experience in full-stack web development and applied deep learning. Experienced in developing scalable web applications using Next.js, TypeScript, and MongoDB, and building machine learning models using TensorFlow and Keras. Strong foundation in data structures, problem-solving, and end-to-end project implementation.

EDUCATION

National Institute of Technology, Kurukshetra Aug 2023 – June 2027
Bachelor of Technology in Artificial Intelligence and Machine Learning

Kendriya Vidyalaya No.1 AFS Pune CBSE Class XII
Percentage: 89.4%

PROJECTS

MNIST Image Generator (GAN) [GitHub Repository](#)

Python, TensorFlow, Keras, NumPy, Matplotlib

- Developed a Generative Adversarial Network (GAN) to synthesize realistic handwritten digits by training on the MNIST dataset of 60,000 images.
- Architected a Generator model to transform random noise into 28x28 grayscale images and a Discriminator model to evaluate image authenticity.
- Optimized model performance using Binary Cross-Entropy loss and the Adam optimizer, achieving stable convergence during training.

Image Captioning System (CNN-LSTM) [GitHub Repository](#)

TensorFlow, Keras, Python

- Built an image captioning model using pretrained Inception V3 for feature extraction and LSTM for sequence generation.
- Preprocessed textual data using tokenization and embeddings on the Flickr8k dataset.
- Evaluated model performance using BLEU-1 and BLEU-2 metrics.

AI vs Real Image Detection System [GitHub Repository](#)

React (TypeScript), Flask, TensorFlow, Keras

- Developed a deep learning-based system to classify images as real or AI-generated using a convolutional neural network.
- Built a React and TypeScript frontend supporting image upload, preview, and prediction results.
- Implemented an image preprocessing pipeline including resizing, normalization, and RGB conversion.
- Exposed trained model inference through RESTful APIs using Flask.

SKILLS

- **Hard Skills:** Machine Learning, Deep Learning, CNN, LSTM, REST API Development, Full-Stack Web Development, Database Design
- **Programming Languages:** JavaScript, TypeScript, Python, C, C++
- **Frameworks & Tools:** React.js, Next.js, Node.js, Express.js, TensorFlow, Keras, Pandas, NumPy, MongoDB, PostgreSQL, SQL
- **Soft Skills:** Problem Solving, Analytical Thinking, Attention to Detail, Self-Learning, Time Management, Team Collaboration

ACHIEVEMENTS AND CERTIFICATIONS

- Solved 500+ Data Structures and Algorithms problems on coding platforms [LeetCode / Codolio](#)
- Completed the Machine Learning Specialization (Regression and Classification) [Coursera Certificate](#)